

CS & IT ENGINEERING

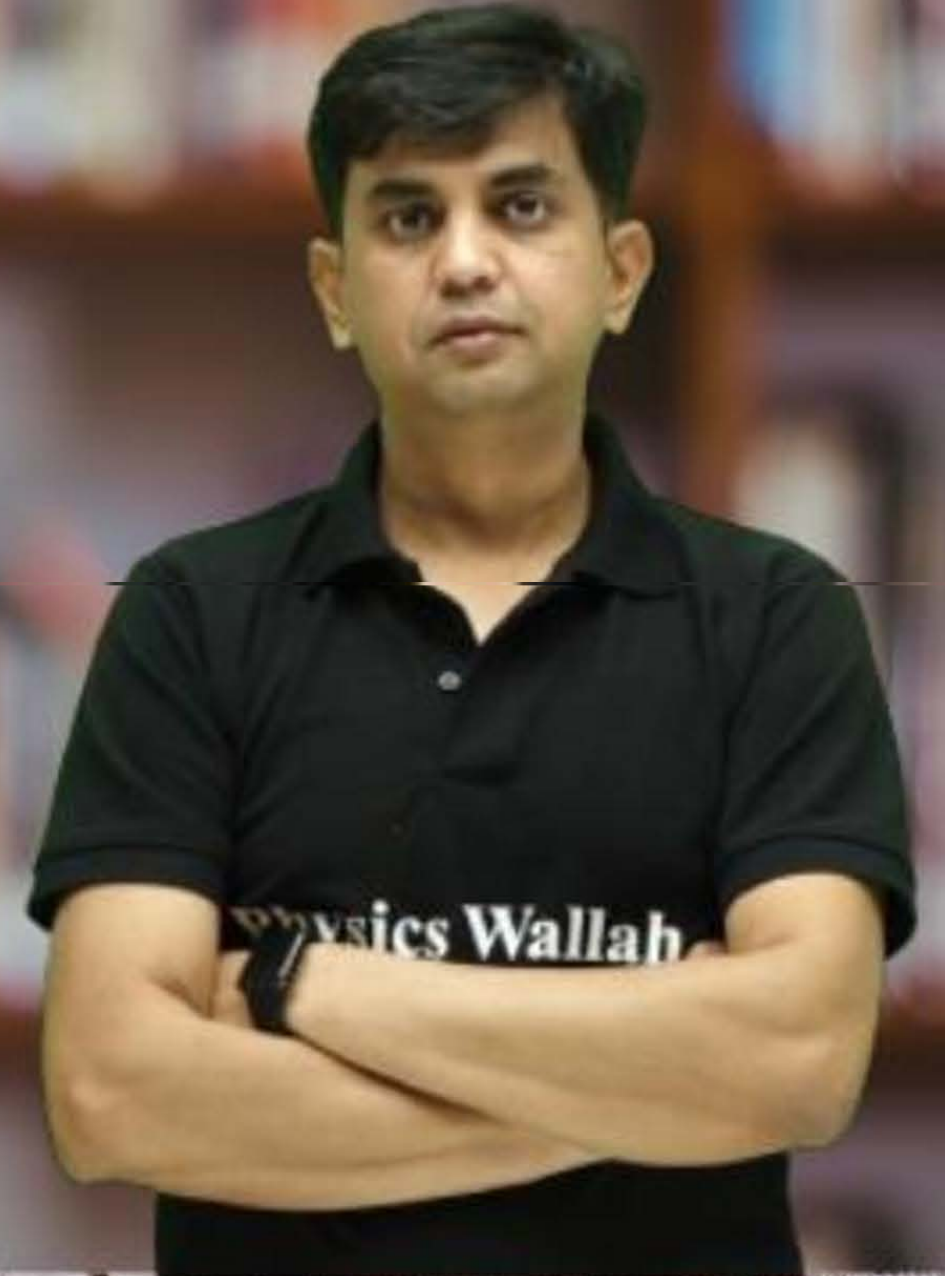


Computer Network-2

MAC Sub-Layer

Lecture No. - 02

By - Abhishek Sir



ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW





Recap of Previous Lecture



Topic

Multiple Access Protocol





Topics to be Covered



Topic

Pure ALOHA

Topic

Slotted ALOHA



Topic : Pure ALOHA



-> When node has packet to transmit, it transmit immediately

← First Time

-> Allow collision to happen
[recover via "retransmission"]

-> Use randomization in choosing "when to retransmit"
[Back-off time]

*



Topic : Pure ALOHA



-> All frames are of equal size

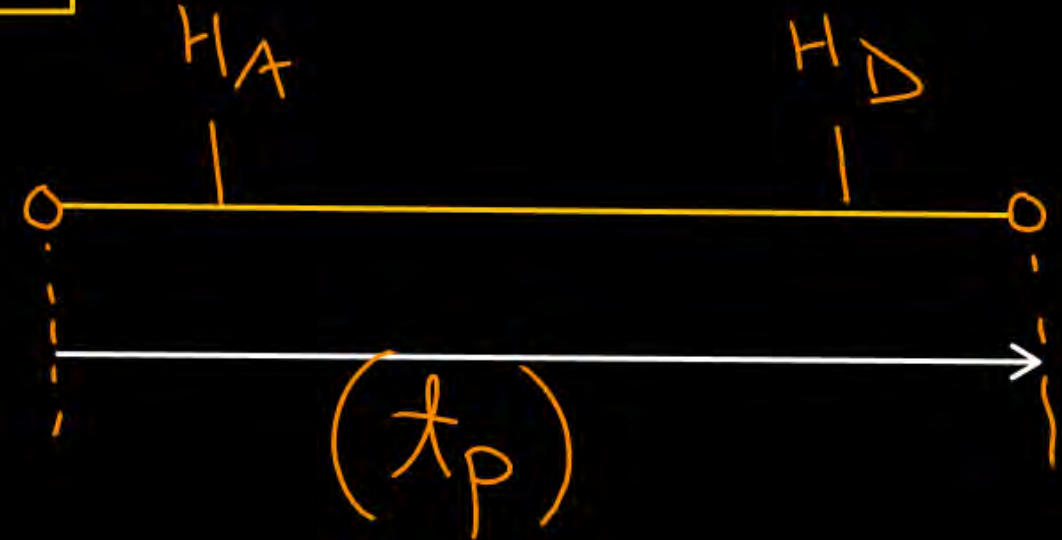
⇒ Assume negligible
propagation delay

-> Receiver send acknowledgement to Source
[for every correctly (without collision) received frame]

-> After transmitting a frame, Source wait for an ACK upto time-out

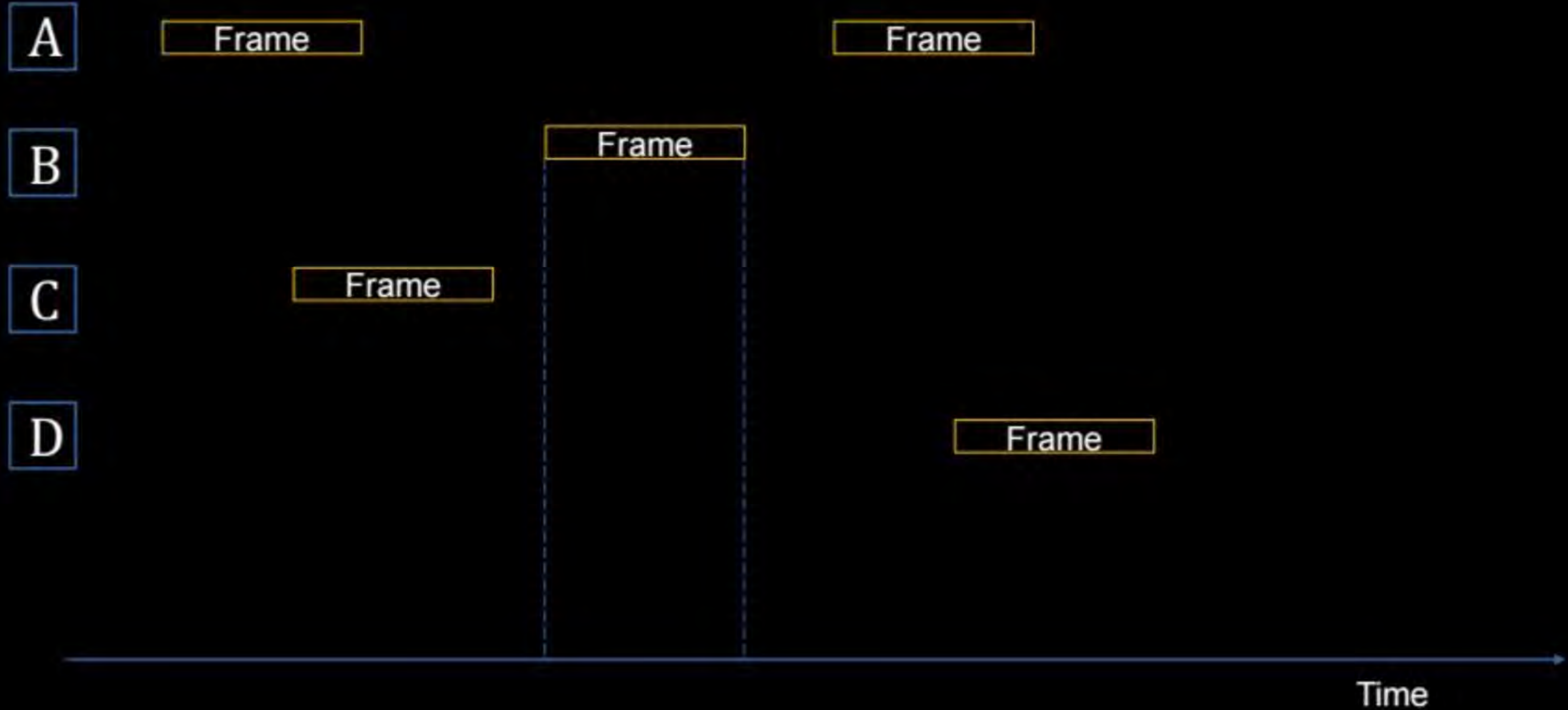
-> Time-out time = Round Trip Propagation Delay

[To value $\geq$ RTT]





Topic : Pure ALOHA





Topic : Pure ALOHA



Vulnerable time :

-> Time duration in which no transmission should be done to avoid collision

Vulnerable time = 2 * Frame Transmission Time



Topic : Pure ALOHA



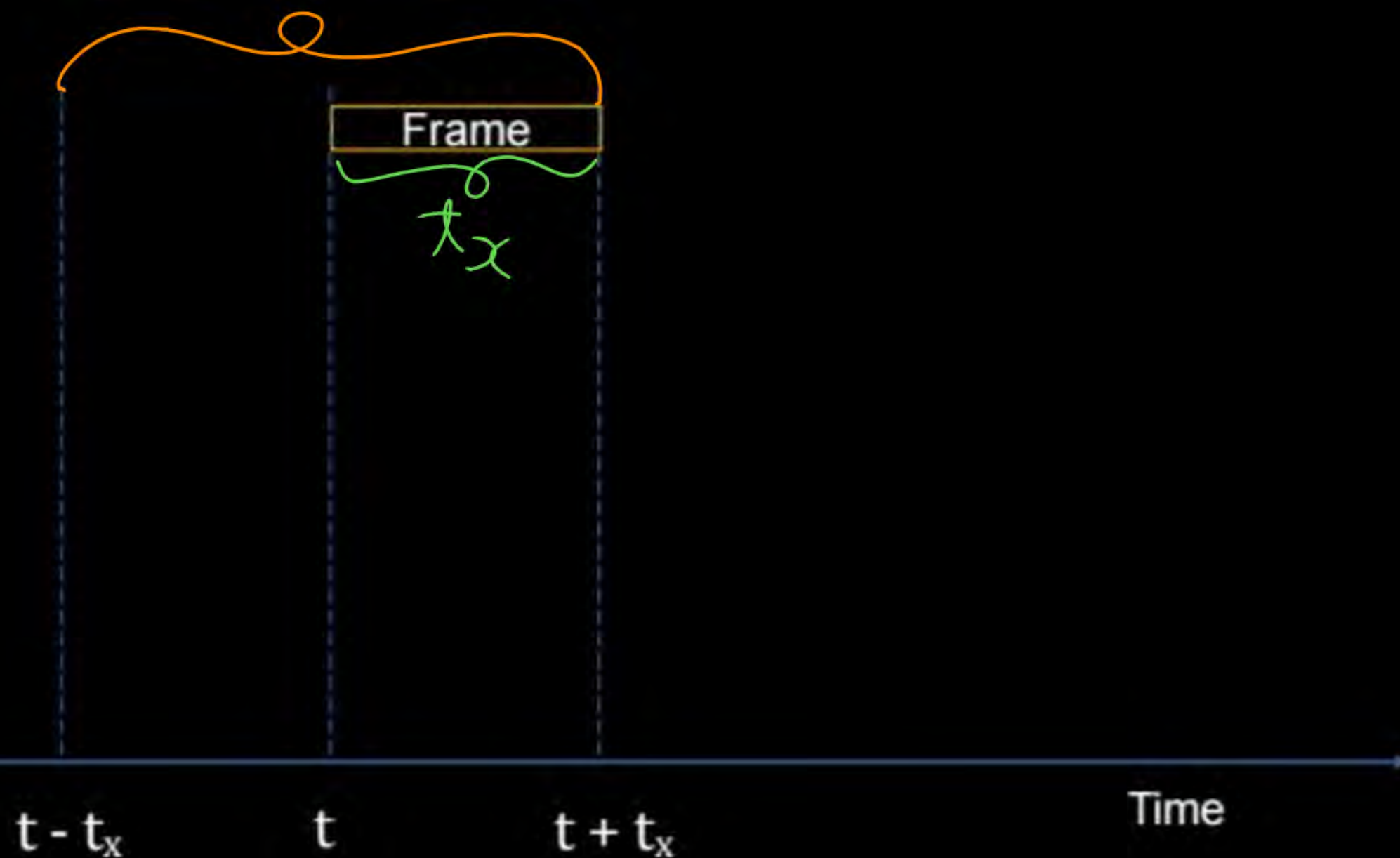
vulnerable time
 $= 2t_x$

A

B

C

D



Example 1 :-

[NAT]



#Q. Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 4 KB (kilo-bytes). The channel transmission rate is 256 Mbps. Calculate vulnerable time (in micro-seconds) ?

Solution 1 :-

$$\text{Frame Size} = 4 \text{ KB} = 2^2 * 2^{10} * 2^3 \text{ bits} = 2^{15} \text{ bits}$$

$$\text{Bandwidth} = 256 \text{ Mbps} = 2^8 * 10^6 \text{ bits/sec}$$

$$\begin{aligned} \text{Frame Transmission Time (t}_x\text{)} &= \frac{\text{Frame Size}}{\text{Bandwidth}} = \frac{2^{15} \text{ bits}}{2^8 * 10^6 \text{ bits/sec}} \\ &= 2^7 * 10^{-6} \text{ sec} = 128 \mu\text{s} \end{aligned}$$

For Pure ALOHA :

$$\begin{aligned} \text{Vulnerable time} &= 2 * \text{Frame Transmission Time} \\ &= 2 * 128 \mu\text{s} = 256 \mu\text{s} \end{aligned}$$

$$\boxed{\text{Ans} = 256}$$



Topic : Pure ALOHA



G : Mean number of transmissions per frame time

[Average number of frame transmitted in one frame transmission time]

S : Throughput per frame time

[Average number of frame transmitted successfully in one frame transmission time]

P = Probability that a frame does not involve in a collision

$$S = G * P$$

$$P = \frac{S}{G}$$

$$S \leq G$$

Example 2 :-

[NAT]



#Q. Consider a network using the pure ALOHA medium access control protocol, where each frame transmission delay is 20 milliseconds. The aggregate number of transmissions (including retransmissions) across all the nodes is 4,000 frames per second. Calculate mean number of frame transmissions per frame time?

$$t_x = 20 \text{ ms}$$

1 sec

→ 4000 frames

10^3 ms

→ 4×10^3 frames

1 ms

→ 4 frames

$(t_x = 20 \text{ ms})$

→ 80 frames

Ans = 80

G = 80



Topic : Pure ALOHA

According to Poisson's distribution

$$P = e^{-2G}$$

$$\underline{S} = \underline{G} * \underline{P}$$

$$S = G * e^{-2G}$$

Vulnerable time = $2 \times$
Mean no. of transmissions
per frame time = G



Example 3 :-

[NAT]



#Q. Consider a network using the pure ALOHA medium access control protocol, where each frame transmission delay is 2 milliseconds. The aggregate number of transmissions (including retransmissions) across all the nodes is 500 frames per second. Calculate throughput per frame time? (Rounded off to three decimal places)

$$t_x = 2 \text{ ms}$$

1 sec $\rightarrow$ 500 frames

10^3 ms $\rightarrow$ 500 frames

1 ms $\rightarrow$ $\frac{1}{2}$ frames

$(t_x = 2 \text{ ms}) \rightarrow$ 1 frame

$$G = 1$$

Pure ALOHA:-

$$S = G * e^{-2G}$$

$$[\text{put } G = 1]$$

$$S = e^{-2} = \frac{1}{e^2} \rightarrow \begin{matrix} 0.13533 \\ 0.13717 \end{matrix}$$

Ans: 0.135 to 0.138

$$P = \frac{S}{G} = \frac{0.135}{1} = 0.135$$

In percent (Throughput)

13.5 % to 13.8 %



[NAT]

IIT-B, H.W.

[GATE-2021][2 Mark]



#Q. Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 1 Mbps ($=10^6$ bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is _____.



Topic : Pure ALOHA



$$S = G * e^{-2G}$$

For what value of G throughput S is maximum ?

$$\frac{dS}{dG} = 0$$



Topic : Pure ALOHA



$$\frac{d}{dG} [G * e^{-2G}] = 0$$

$$G * \frac{d}{dG} e^{-2G} + e^{-2G} * \frac{d}{dG} G = 0$$

$$G * e^{-2G} * (-2) + e^{-2G} = 0$$

$$e^{-2G} - 2G * e^{-2G} = 0$$

$$e^{-2G} * [1 - 2G] = 0$$

$$1 - 2G = 0$$

$$G = \frac{1}{2}$$



Topic : Pure ALOHA



$$S = G * e^{-2G}$$

Maximum throughput (S) can achieve at $G = \frac{1}{2}$

$$[\text{put } G = \frac{1}{2}]$$

$$S = \frac{1}{2} * e^{-2 * \frac{1}{2}}$$

$$S = \frac{1}{2e}$$



Topic : Pure ALOHA



$$S = G * e^{-2G}$$

Maximum throughput (S) can achieve at G = 0.5 = $\frac{1}{2}$

$$\text{Maximum throughput} = \boxed{\frac{1}{2e}} = \underbrace{0.1839} = \boxed{18.39\%}$$

↓

$$S = 0.1839$$

$$G = \frac{1}{2}$$

$$P = \frac{S}{G}$$



2 mins Summary



Topic

Pure ALOHA ✓

Topic

~~Slotted ALOHA~~



THANK - YOU

